A Seminar Presentation
On

Electron Spin Resonance (ESR)

OR

Electron Paramagnetic Resonance (EPR)

By Kumbhar Kaustubh Rajshekhar

ESR Spectroscopy:

It is based upon the absorption of radio wave by an unpaired electron when exposed to a strong magnetic field.

It is shown by

1.Atoms having an odd number of electrons.

2.Ions having partly filled inner orbitals.

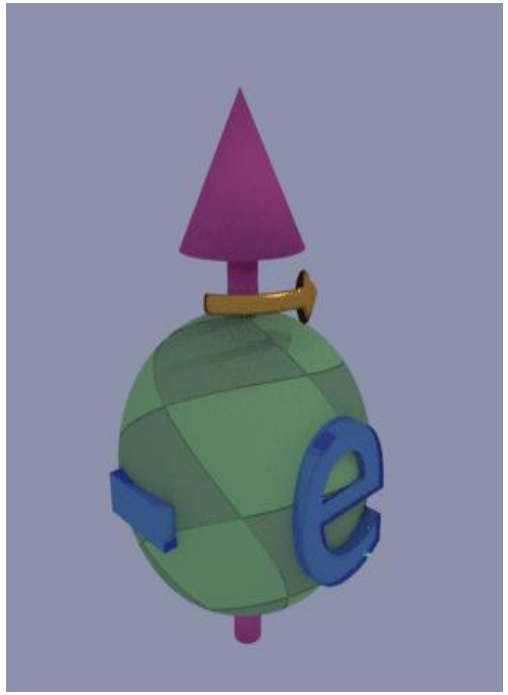
3.Free radicals having unpaired electron.

Electron Spin :

An electron spin is a property of an electron that is related to its spin about an axis of its angular momentum.

It is one of the three intrinsic properties of an electron.

$+\frac{1}{2}$

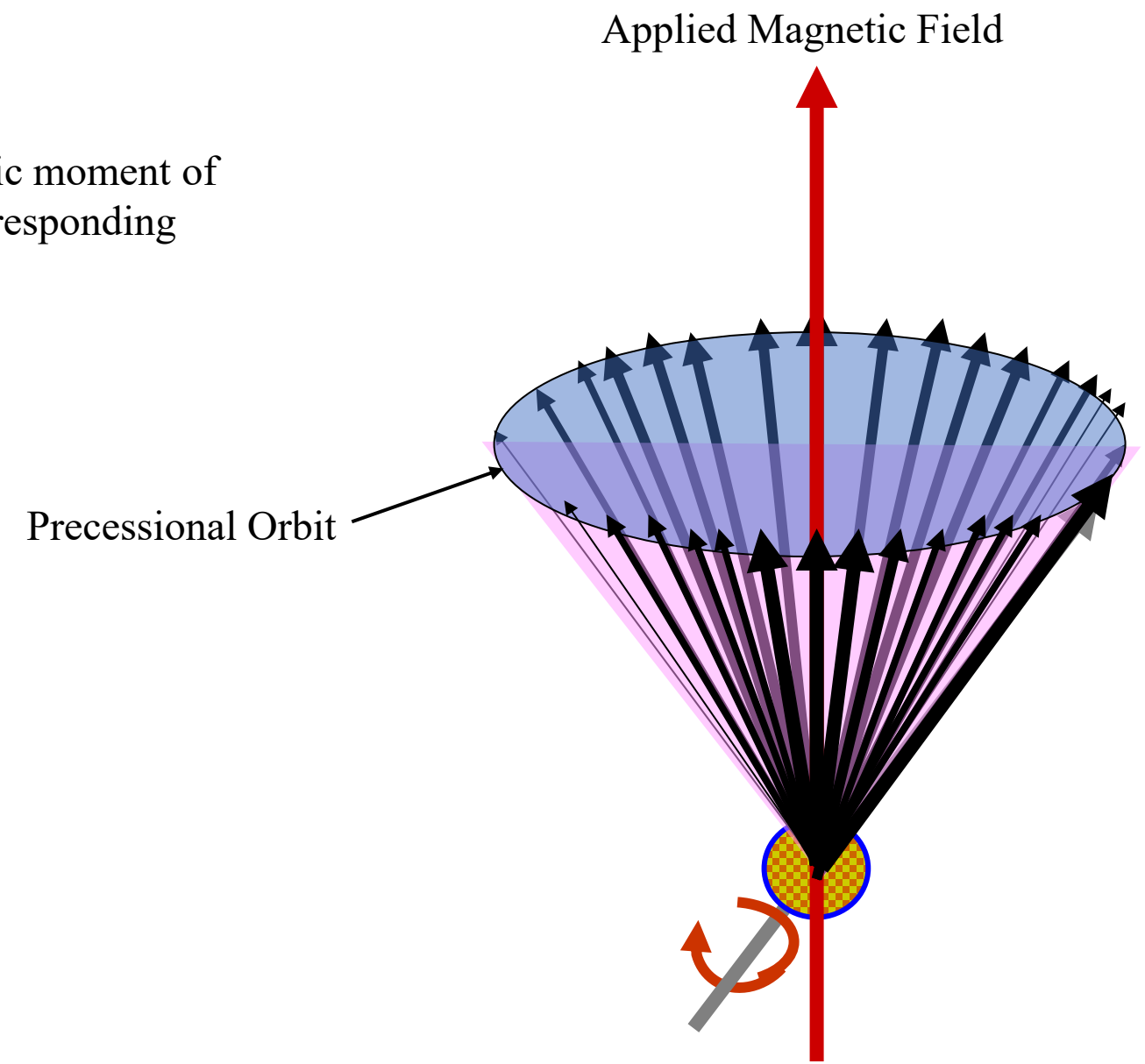


$-\frac{1}{2}$

Lamour Precession :

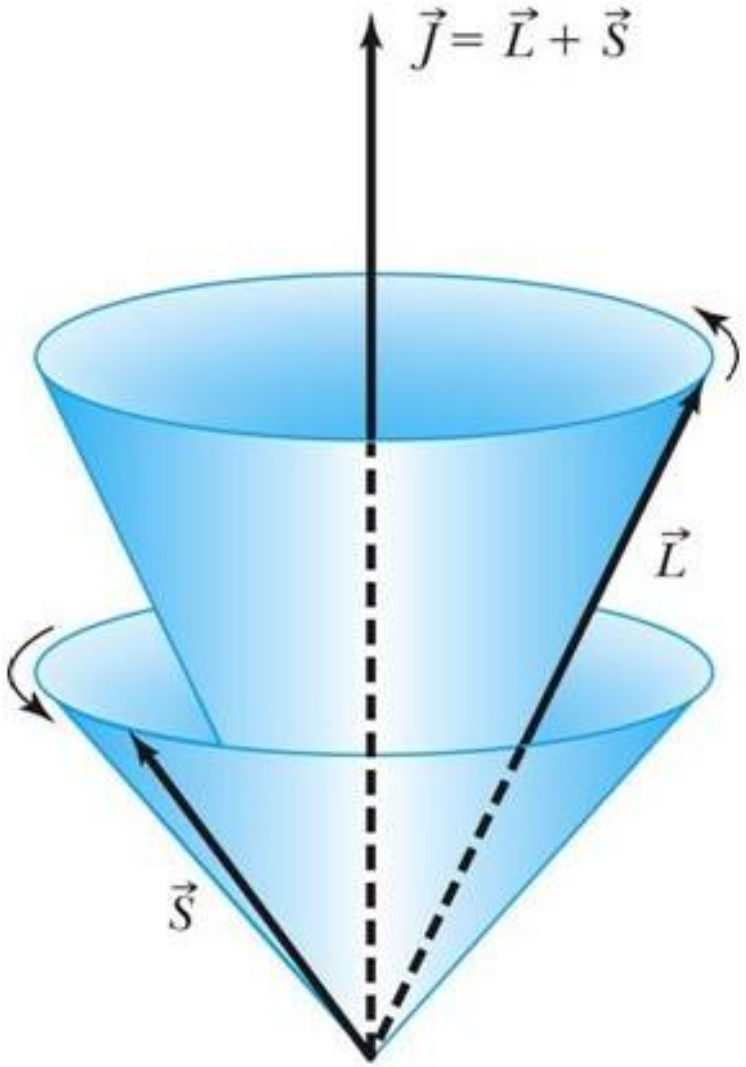
Larmor precession is the precession of the magnetic moment of an object about an external magnetic field and corresponding frequency is called Larmor frequency.

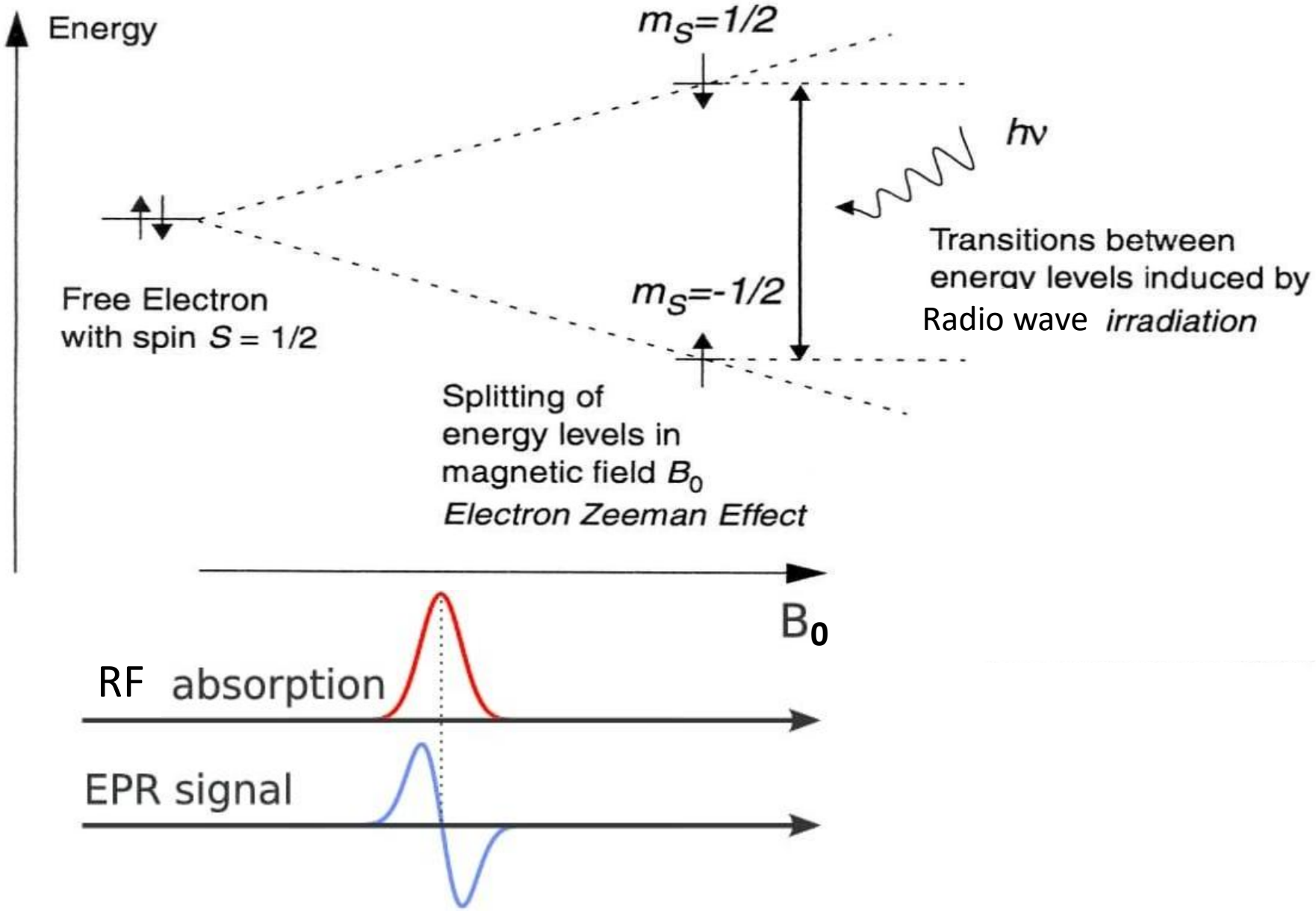
$$\nu_0 = g\left(\frac{\mathrm{e}}{4\pi m}\right)\mathrm{H}$$



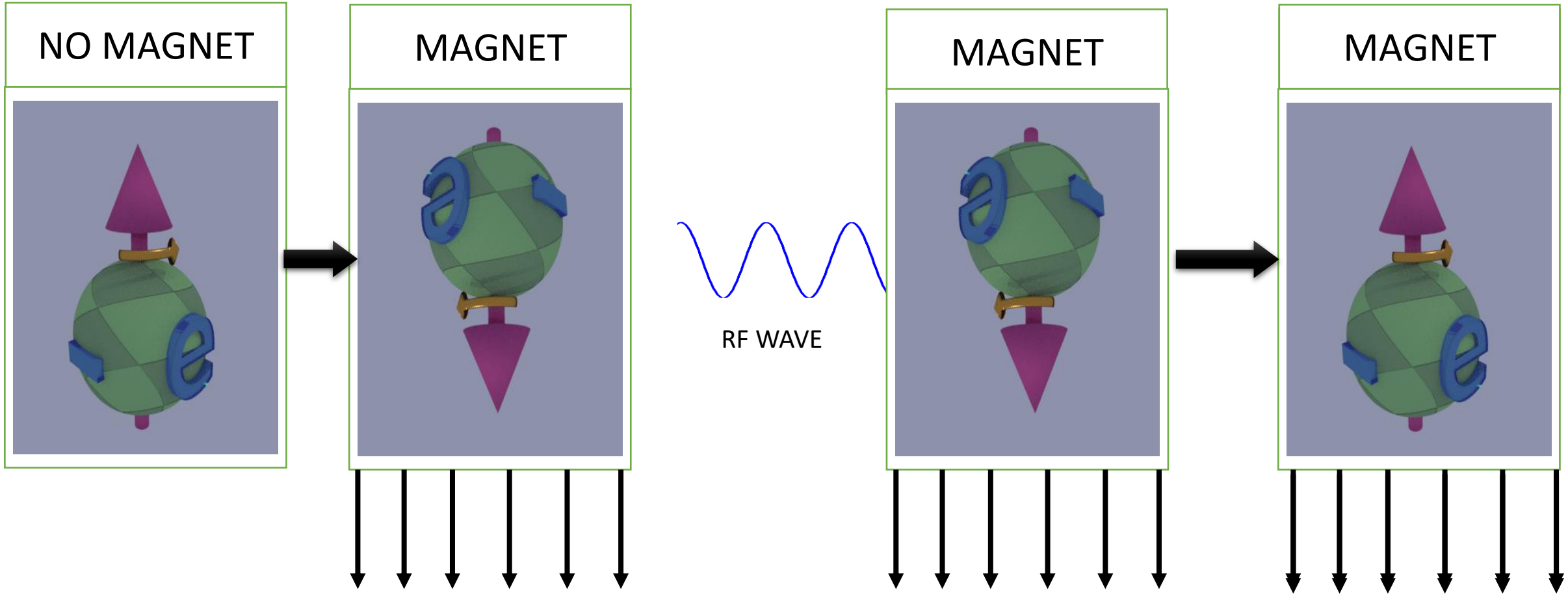
L-S Coupling:

The spin angular momentum of electron S couples with the orbital angular momentum of the electron L to give a resultant J and this coupling gives the fine structure of the spectra.




Energy
m_S=1/2
hν
Transitions between
energy levels induced by
Radio wave irradiation
Free Electron
with spin S = 1/2
m_S=-1/2
Splitting of
energy levels in
magnetic field B_0
Electron Zeeman Effect
B_0
RF absorption
EPR signal

ESR:



Aim:

To determine Lande's splitting factor (g) by ESR spectrometer.

Apparatus:

Radio Frequency Oscillator, ESR Spectrometer, Helmholtz Coil, Sample- DPPH (Di Phenyl Picryl Hydroxyl) etc.

Formula:

1. $g = \frac{h\nu_0}{\mu_0 H_0}$

where, g = Lande's splitting factor

h = Planks constant = 6.625×10^{-27} erg.sec

ν_0= Resonant frequency in Hz

μ_0 = Bohr magneton = 0.927×10^{-20} erg/gauss

H_0 = Magnetic field on sample at resonance in gauss

2. $H_0 = H_{pp} \cdot \frac{qI}{p}$

where, Hpp = 165.25gauss/amp

qI = Slope of graph

p= Distance between two bases

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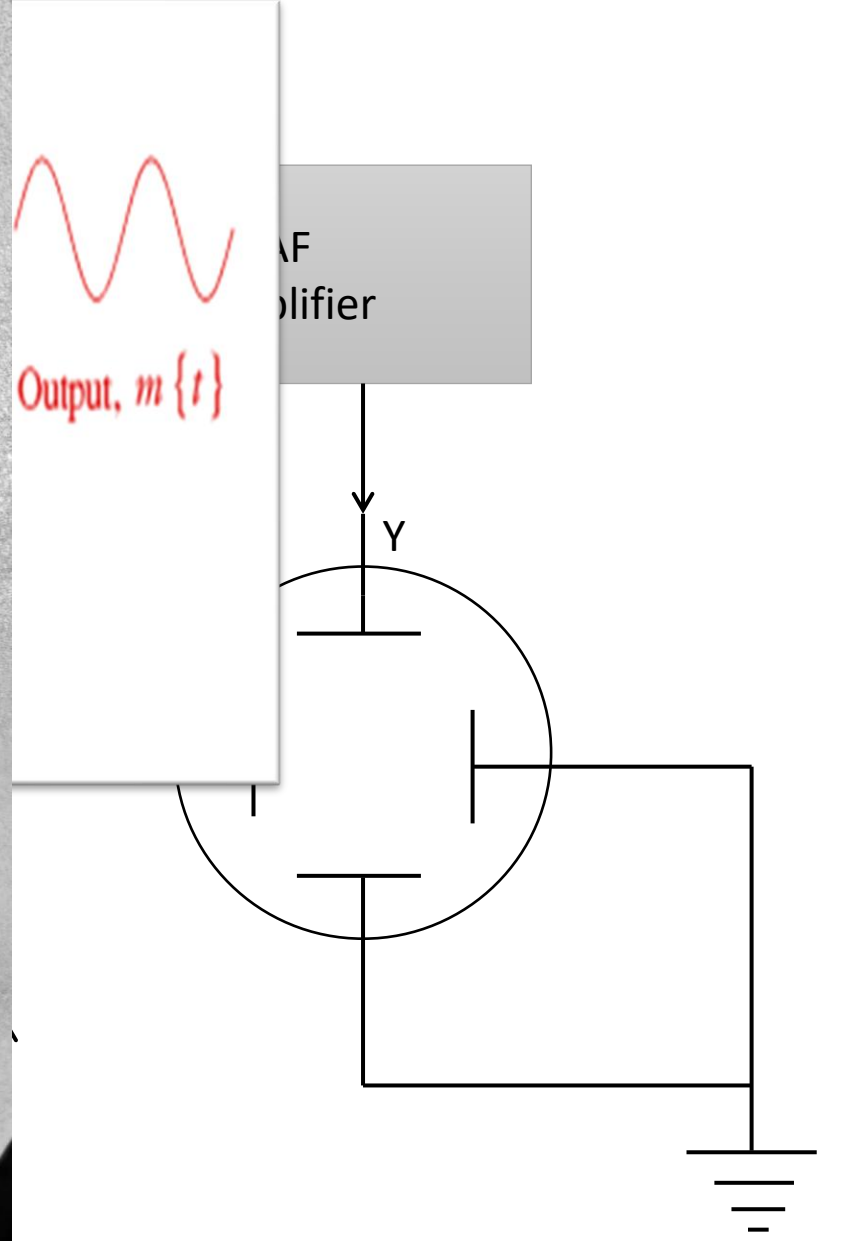
ROORKEE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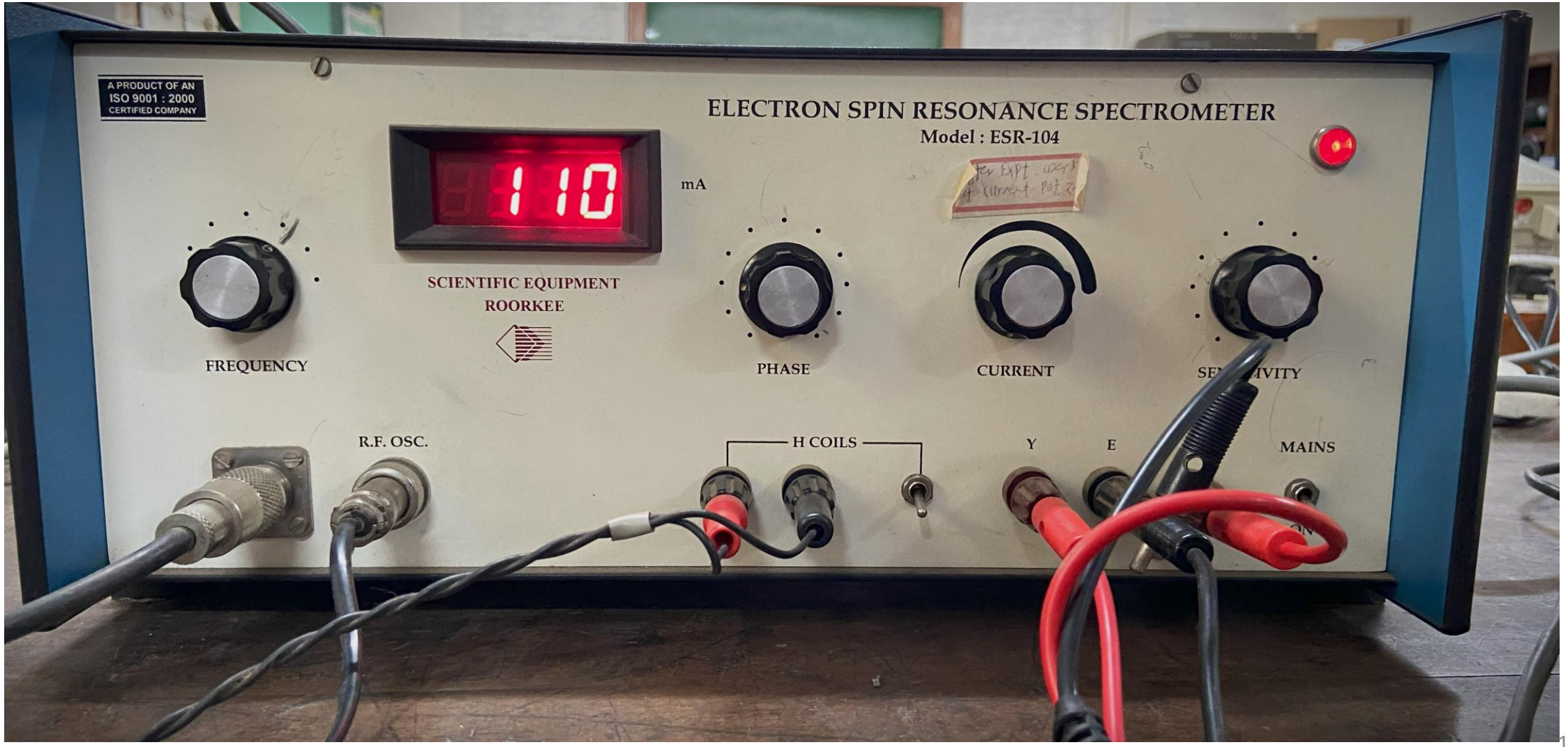
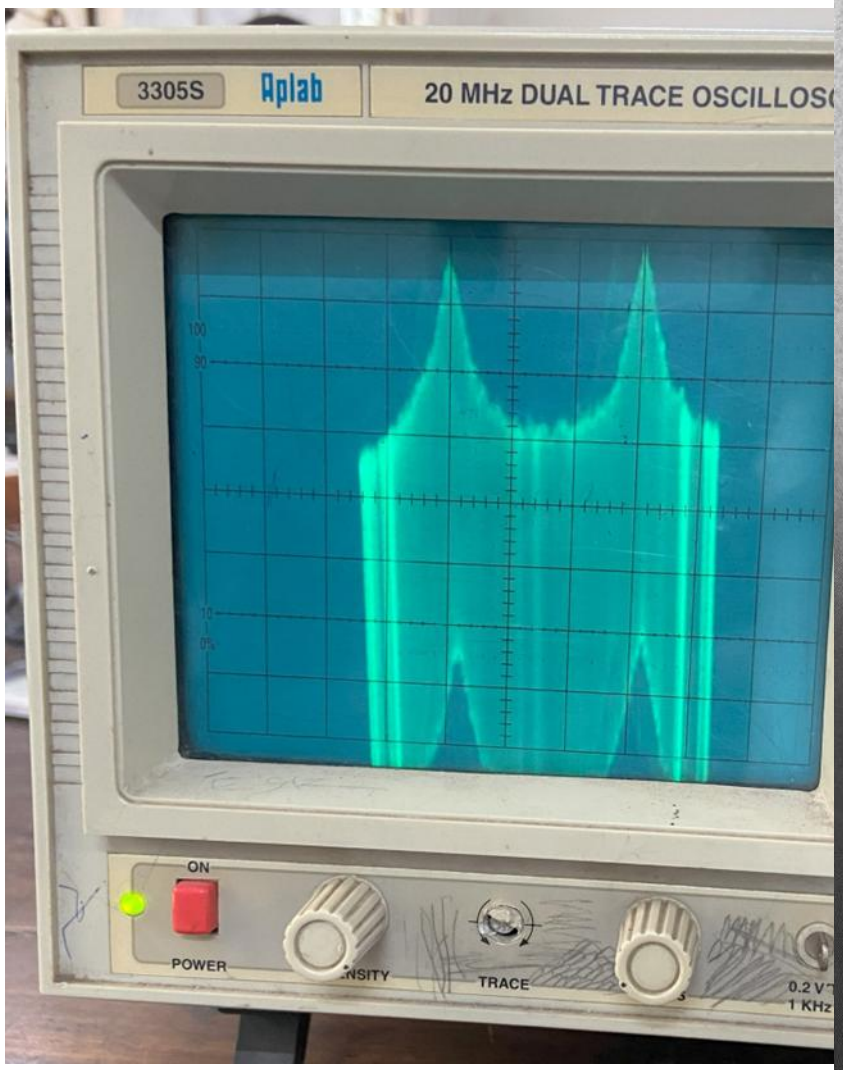
Actual Experimental Setup:




3305S
Aplab
20 MHz DUAL TRACE OSCILLOS
ON
POWER
TRACE
0.2 V
1 KHz

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MHz
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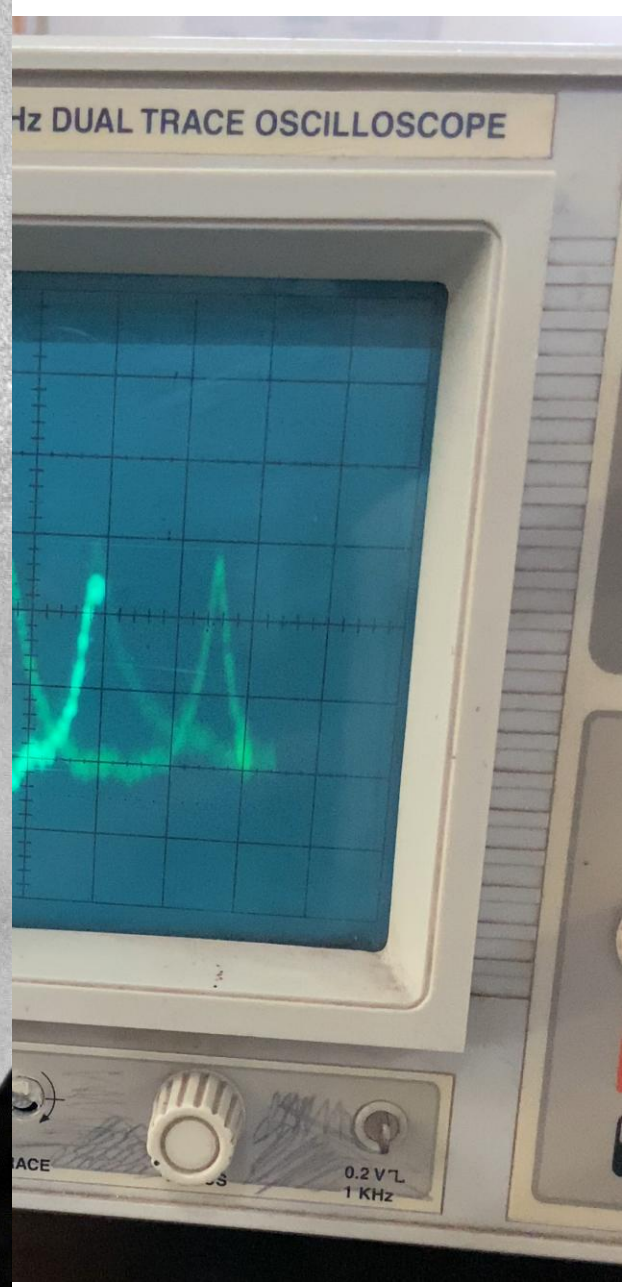
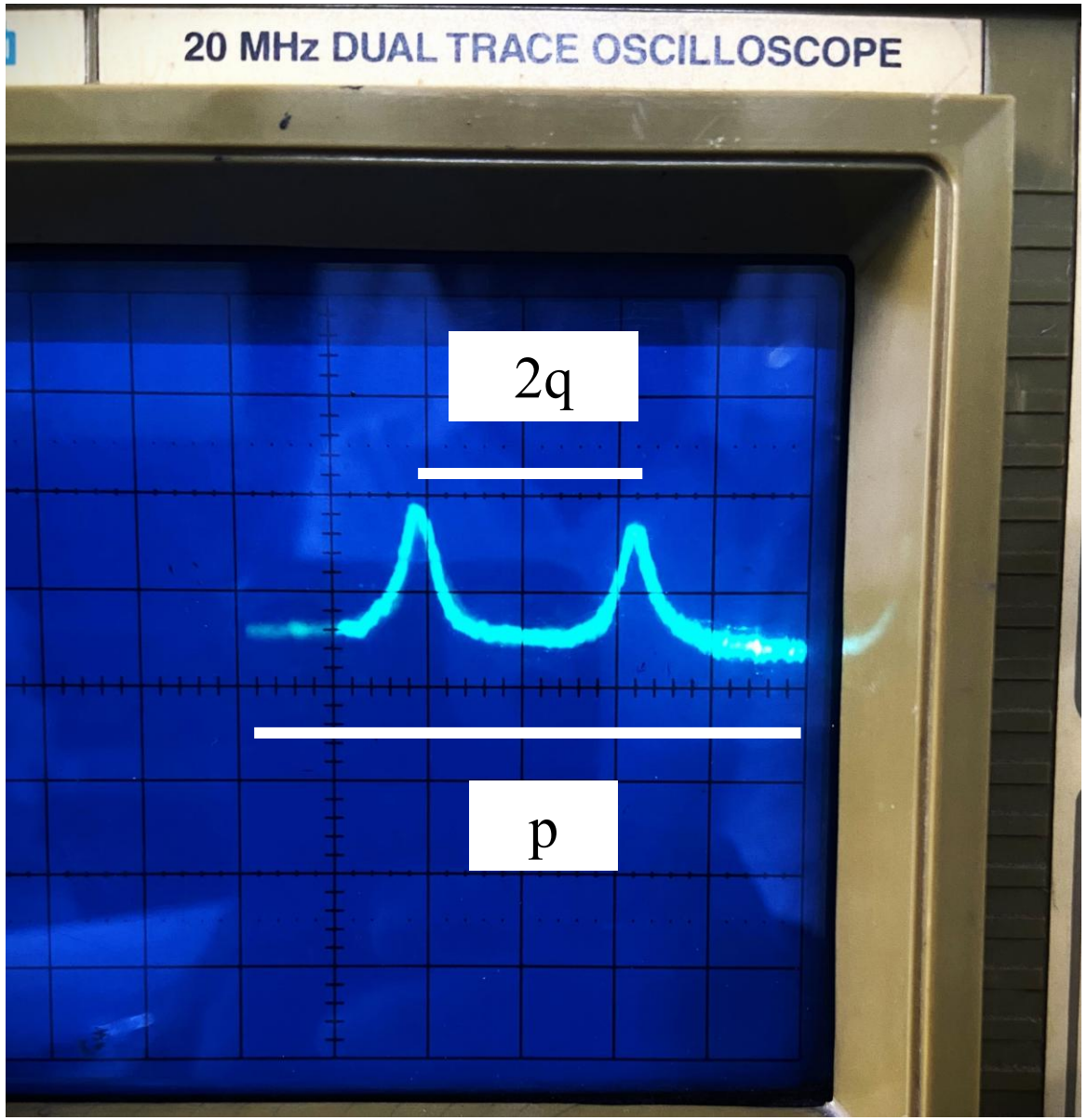

Hz DUAL TRACE OSCILLOSCOPE
0.2 V
1 KHz

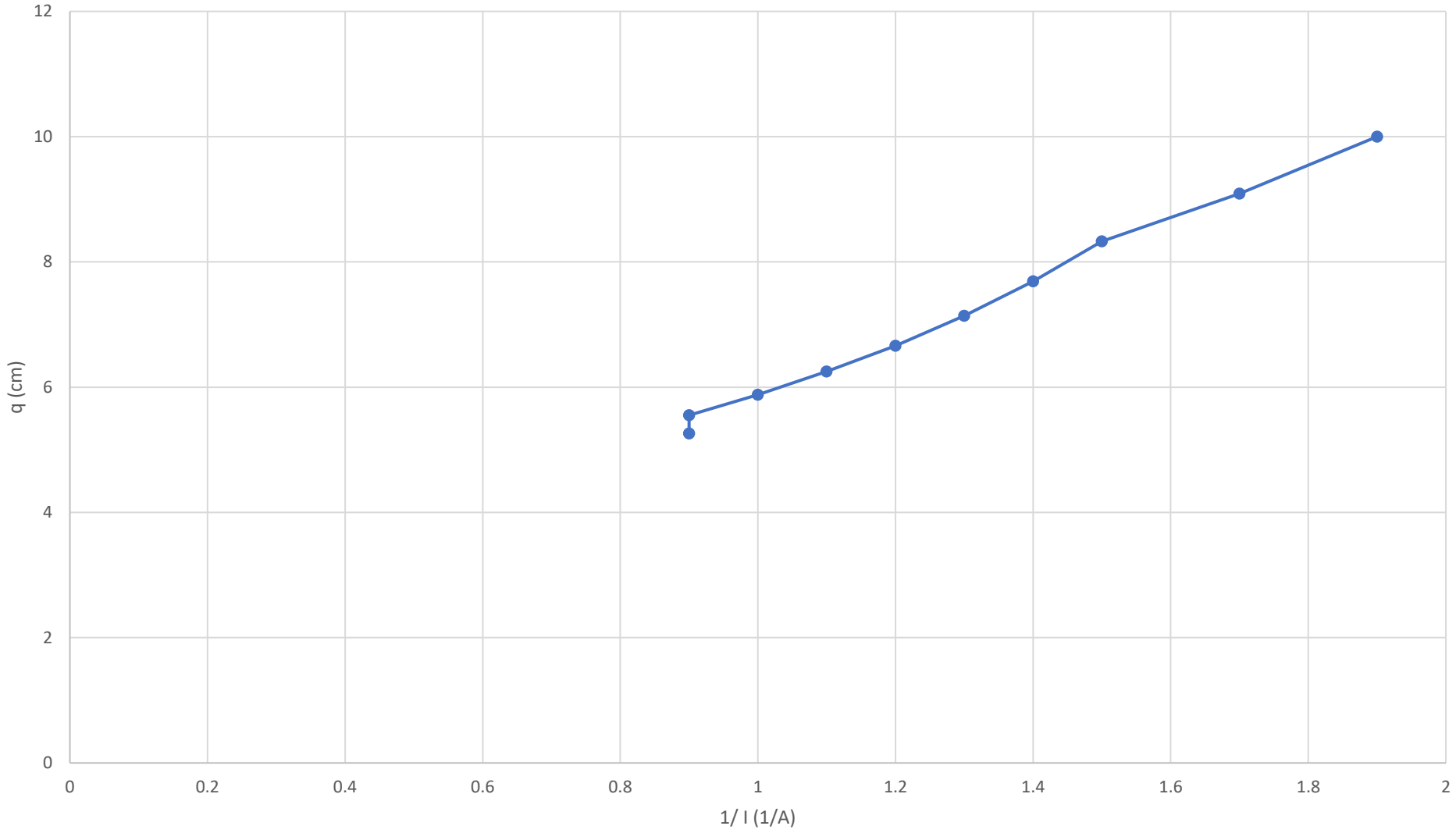
Resonance Peaks:



Observation Table:

Obs. No.	Current 'I' (mA)	Distance between peaks '2q' (cm)	'q' (cm)	Current 'I' (Amp)	1/I (A^-1)	q.I
1	100	3.8	1.9	0.10	10	0.19
2	110	3.4	1.7	0.11	9.09	0.18
3	120	3	1.5	0.12	8.33	0.18
4	130	2.8	1.4	0.13	7.69	0.182
5	140	2.6	1.3	0.14	7.14	0.18
6	150	2.4	1.2	0.15	6.66	0.176
7	160	2.2	1.1	0.16	6.25	0.17
8	170	2	1	0.17	5.88	0.17
9	180	1.8	0.9	0.18	5.55	0.162
10	190	1.8	0.9	0.19	5.26	0.171

Graph:



Result:

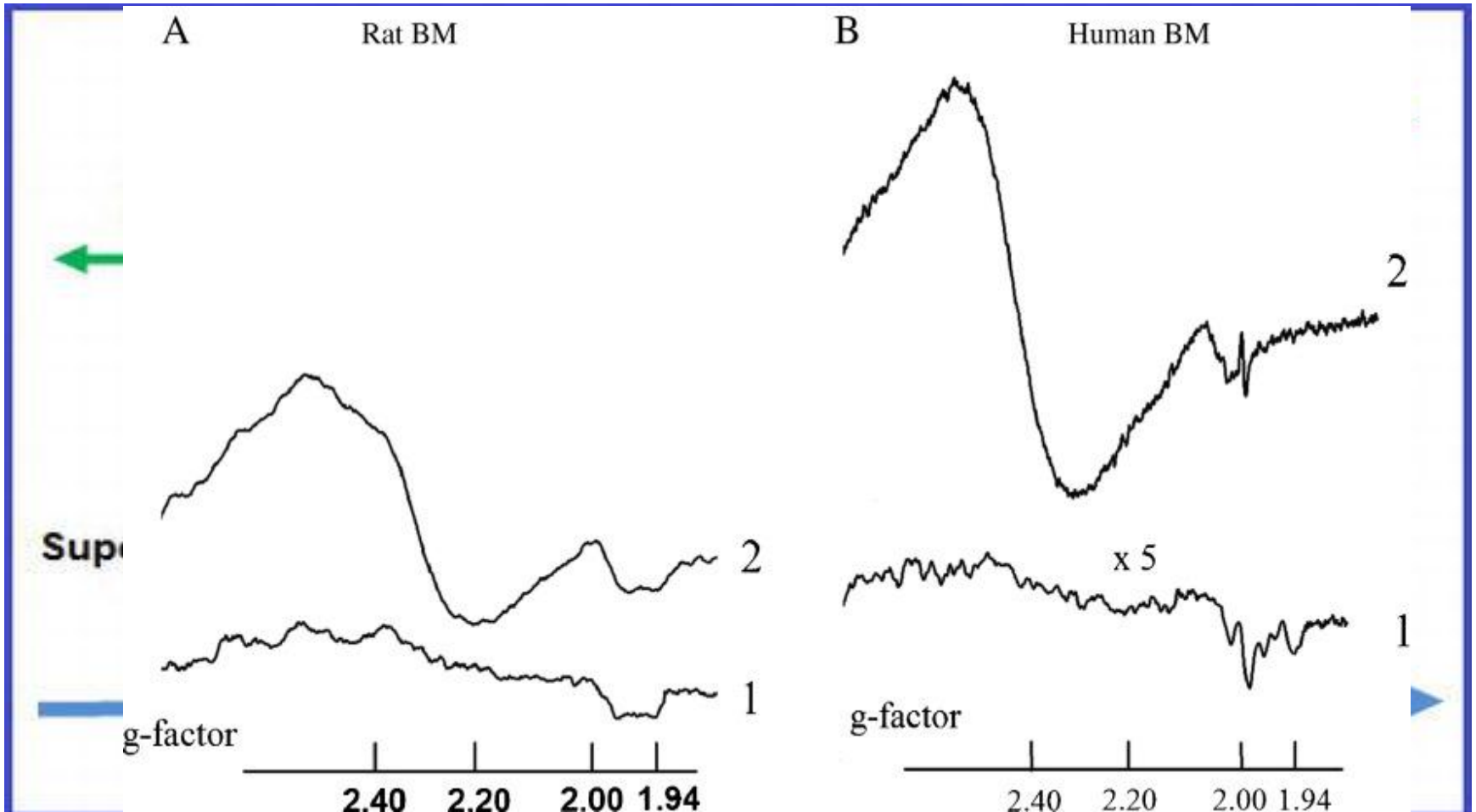
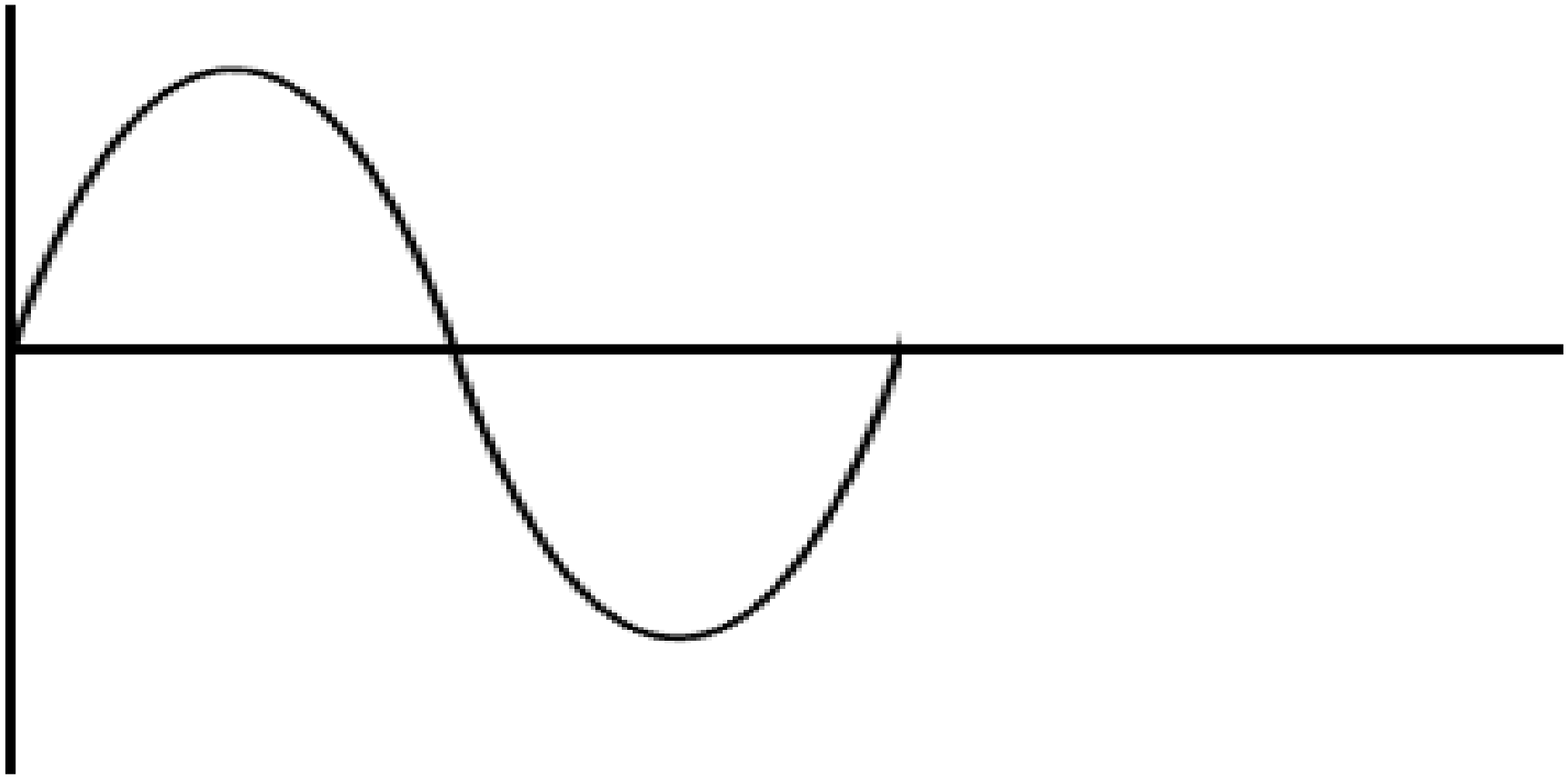
The value of ‘g’ ESR Spectrometer Method is 2.2.

Conclusion:

When current increases, the distance between peaks decreases.

Application:

1. It is the <u>ONLY</u> method in which direct detection of free radicals is possible.
2. Dignosis of human cancer cells.
3. In research and development of Biomedicine (MRI Agent & Drug Delivery).
4. Identification of irradiated food (as well as poisoned)



Thank You